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TREND-BASED MOCK PAPER

GATE Agricultural Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Agricultural Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Area of rectangle 10×5 :

- (A) 15
- (B) 50
- (C) 25
- (D) 100

Section B: Agricultural Engineering

Q2. [1-NAT] [1 mark]

Derivative of e^x :

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2

- (B) 0.5
- (C) 0.6
- (D) 0.4

Q6. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q7. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q8. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q9. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q10. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q11. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q12. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2, 1 \\ 1, 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q13. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q14. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q15. [1-MCQ] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q16. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q17. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q19. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q21. [2-MSQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi \cdot r^2$

- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q22. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q23. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q24. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q25. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q26. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q28. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q29. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q31. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2, 1 \\ 1, 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
 - (B) $[2, 1]^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q33. [1-MCQ] [1 mark]

Primary tillage implements:

- (A) Harrow
 - (B) Plough
 - (C) Seeder
 - (D) Sprayer
-

Q34. [1-NAT] [1 mark]

Tractor power measured in:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
 - (B) HP
 - (C) Both kW and HP
 - (D) Torque only
-

Q36. [2-MCQ] [2 marks]

Primary tillage implements:

- (A) Harrow
 - (B) Plough
 - (C) Seeder
 - (D) Sprayer
-

Q37. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW

- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q38. [2-NAT] [2 marks]

Primary tillage implements:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q40. [2-MCQ] [2 marks]

Primary tillage implements:

- (A) Harrow
- (B) Plough
- (C) Seeder
- (D) Sprayer

Q41. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q42. [2-NAT] [2 marks]

Primary tillage implements:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q44. [1-MCQ] [1 mark]

Most abundant element in Earth's crust:

- (A) Iron
- (B) Oxygen
- (C) Silicon
- (D) Aluminum

Q45. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure

- (C) Water only
- (D) Air only

Q46. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q47. [2-MCQ] [2 marks]

Most abundant element in Earth's crust:

- (A) Iron
- (B) Oxygen
- (C) Silicon
- (D) Aluminum

Q48. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q49. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q50. [2-MCQ] [2 marks]

Most abundant element in Earth's crust:

- (A) Iron
- (B) Oxygen
- (C) Silicon
- (D) Aluminum

Q51. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q52. [1-MCQ] [1 mark]

Photosynthesis equation: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose} +$

- (A) O_2
- (B) CO_2
- (C) H_2
- (D) N_2

Q53. [1-MCQ] [1 mark]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
- (B) Rubisco
- (C) NADPH
- (D) ATP only

Q54. [2-MCQ] [2 marks]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
- (B) Rubisco
- (C) NADPH
- (D) ATP only

Q55. [2-MSQ] [2 marks]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
- (B) Rubisco
- (C) NADPH
- (D) ATP only

Q56. [2-MCQ] [2 marks]

Photosynthesis equation: CO₂ + H₂O → glucose +:

- (A) O₂
- (B) CO₂
- (C) H₂
- (D) N₂

Q57. [2-MCQ] [2 marks]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
- (B) Rubisco
- (C) NADPH
- (D) ATP only

Q58. [1-MCQ] [1 mark]

4-stroke engine has:

- (A) 2 strokes
- (B) 4 strokes per cycle
- (C) 1
- (D) 3

Q59. [1-MCQ] [1 mark]

Tractor power measured in:

- (A) HP
- (B) kW
- (C) Both
- (D) None

Q60. [1-MCQ] [1 mark]

Drip irrigation saves:

- (A) 30%
- (B) 50%
- (C) 70%
- (D) 90%

Q61. [1-MCQ] [1 mark]

Field capacity is:

- (A) Saturation
- (B) Optimal water
- (C) Wilting point
- (D) Dry

Q62. [1-NAT] [1 mark]

Fumigation controls:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Pasteurization temperature:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-NAT] [1 mark]

Chain survey used for:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Solar panel converts:

- (A) Heat to electric
 - (B) Light to electric
 - (C) Wind to electric
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	A	1
3	1MCQ	B	1
4	1MCQ	C	1
5	1MCQ	B	1
6	1MCQ	B	1
7	1NAT	A	1
8	1NAT	A	1
9	1NAT	B	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1NAT	B	1
15	1MCQ	C	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1NAT	C	1
19	1MCQ	B	1
20	1MCQ	A	1
21	1MCQ	C	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1MCQ	C	1

29	1MCQ	B	1
30	1MCQ	A	1
31	2NAT	A	2
32	2NAT	A	2
33	2NAT	A	2
34	2MCQ	C	2
35	2MSQ	A	2
36	2MCQ	A	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	C	2
40	2NAT	B	2
41	2MCQ	A	2
42	2MCQ	C	2
43	2MCQ	C	2

44	2NAT	A	2
45	2MCQ	B	2
46	2MCQ	B	2
47	2MCQ	B	2
48	2MCQ	B	2
49	2MCQ	B	2
50	2MCQ	C	2
51	2MCQ	B	2
52	2NAT	A	2
53	2NAT	B	2
54	2MSQ	B	2
55	2MCQ	A	2
56	2MCQ	B	2
57	2MCQ	C	2
58	2MCQ	A	2

59	2MCQ	B	2
60	2MCQ	B	2
61	2MCQ	B	2
62	2MCQ	C	2
63	2MCQ	B	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Farm Machinery & Power | Topic: Engines | Difficulty: difficult

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Farm Machinery & Power | Topic: Tractors | Difficulty: difficult

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Irrigation & Drainage | Topic: Water Management | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Soil and Water Conservation | Topic: Soil Mechanics | Difficulty: difficult

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Post Harvest Engineering | Topic: Storage | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Food Processing | Topic: Preservation | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Leveling | Topic: Land | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Renewable Energy | Topic: Solar | Difficulty: easy

Q11.

Answer: A

Solution:

$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q12.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q13.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q14.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q17.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: difficult

Q19.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO_2 in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: easy

Q20.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q21.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q24.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1) \approx 0.68$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q34.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q36.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q43.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q46.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q47.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q48.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q49.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q51.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q52.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q54.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q55.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q57.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q58.

Answer: A

Solution:

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: difficult

Q59.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: easy

Q60.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q61.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q62.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q63.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO_2 in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: difficult

Q64.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q65.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 31 questions, 47 marks
Farm Machinery: 10 questions, 18 marks
Soil Science: 9 questions, 16 marks
Crop Physiology: 6 questions, 10 marks
Farm Machinery & Power: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Irrigation & Drainage: 1 questions, 1 marks
Soil and Water Conservation: 1 questions, 1 marks
Post Harvest Engineering: 1 questions, 1 marks
Food Processing: 1 questions, 1 marks
Surveying & Leveling: 1 questions, 1 marks
Renewable Energy: 1 questions, 1 marks

Difficulty Distribution:

Easy: 21 | Moderate: 32 | Difficult: 12

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.